Dr. Dennis Ulbrich

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Research Profile

Mathematics: Nonlinear PDEs, Dynamical Systems, Kinetic Theory

My research focuses on nonlinear partial differential equations, dynamical systems, and kinetic theory, with an emphasis on the long-time behavior of complex systems. A central theme of my recent work was the use of entropy methods and hypocoercivity techniques to establish convergence to equilibrium and stability properties.

I have contributed to the analysis of nonlinear wave phenomena and excitation dynamics, as well as to interacting particle systems such as cellular automata.

More recently, I was a postdoctoral researcher at the University of Münster (Institute for Analysis and Numerics) where my work has shifted towards kinetic equations, in particular the Boltzmann equation, its BGK-type approximations, and discrete hypocoercivity methods in collaboration with Prof. Marlies Pirner.

My research combines analytical techniques from PDE theory, ergodic theory, and dynamical systems, with the aim of understanding qualitative and quantitative properties of nonlinear evolution equations.

Publications

Peer-reviewed Articles

1. A. Pauthier, J.D.M. Rademacher, **D. Ulbrich**
   Weak and strong interaction of excitation kinks in scalar parabolic equations
   J Dyn Diff Equat35, 2199–2235, (2023) [DOI] [arXiv]

2. M. Keßeböhmer, J.D.M. Rademacher, **D. Ulbrich**
   Dynamics and topological entropy of 1D Greenberg-Hastings cellular automata
   Ergodic Theory and Dynamical Systems, 41 (2020), no. 5, 1397–1430 [DOI] [arXiv]

Dissertation

D. Ulbrich
Ergodic theory of nonlinear waves in discrete and continuous excitable media
PhD thesis [DOI]

Work in Progress

- Discrete hypocoercivity for a nonlinear kinetic reaction model without initial close-to-equilibrium assumption (with M. Pirner, L. Liu)

Employment

Private tutoring 05/2025 – present

Research Assistant, Hochschule Bremen 02/2025 – 05/2025

- Mathematical modeling and analysis for AI-based handwritten text recognition
- Application of quantitative methods to real-world data problems

Lecturer in Mathematics, Hochschule Bremen 10/2024 – 03/2025

Postdoctoral Researcher, Universität Münster, 06/2023 – 06/2024
Institute for Analysis and Numerics, supervisor: M. Pirner

- Research on discrete hypocoercivity within DFG 456849348
- Development and analysis of discrete hypocoercivity methods
- Collaboration in an interdisciplinary and international research environment

IT Consultant (Internship), Affinis AG 09/2022 – 02/2023

Lecturer in Mathematics, Jacobs University Bremen 01/2022 – 06/2022

Research Assistant (PhD), University of Bremen 04/2017 – 09/2021
Supervisors: J. Rademacher, M. Keßeböhmer

- Research within DFG 384027439
- Research on dynamical systems and nonlinear wave equations

Teaching Experience

Mathematik 1, Hochschule Bremen 10/2024 – 03/2025

- Full course responsibility (lectures, exercises, examinations)
- Design and coordination of course structure and assessments

Finite Mathematics, Jacobs University Bremen 01/2022 – 06/2022

- Lectures and tutorials for interdisciplinary student groups
- Preparation and grading of exams

Introduction to Dynamical Systems, Jacobs University Bremen 01/2022 – 06/2022

- Proseminar and supervision of student presentations

Education

Dr. rer. nat. Mathematics, University of Bremen 10/2017 – 09/2021

Brückenstipendium, University of Bremen 10/2016 – 03/2017

M.Sc. & B.Sc. Mathematics, University of Bremen 10/2011 – 09/2016

Academic Service

- Supervision of undergraduate students
- Organization of academic events and schools

Skills

Matlab, Mathematica, Java, C++, SQL: Basic
LaTeX: Advanced
Languages: German (native), English (fluent)

CV version of March 2026